Causality Assessment in Pharmacovigilance

Overview

Causality assessment in pharmacovigilance is the process of evaluating the relationship between a suspect drug and an adverse event.

1. Core Categories of Causality Assessment

Clinical Factors

- **Temporal Relationship:** Did the adverse event occur within a plausible timeframe after starting the drug?
- **De-challenge:** Did the event improve or resolve when the drug was stopped or the dose reduced?
- **Re-challenge:** Did the event recur if the drug was re-administered? (*Note: Re-challenge is rare due to ethical/safety concerns*).

Drug-Related Factors

- **Labeledness / Signal:** Is the event a known side effect in the package insert or product label?
- **Concomitant Medications:** Could another drug the patient is taking explain the reaction or interact with the suspect drug?

Patient-Related Factors

- **Medical History:** Could an underlying disease or prior condition cause the event?
- **Pharmacogenomics:** Do genetic factors (e.g., specific CYP enzyme variants) influence drug metabolism or susceptibility?

Event-Related Factors

- **Objective Evidence:** Are there clinical lab tests, biopsies, or imaging confirming the nature of the reaction?

2. Standard Methodologies: Naranjo vs. WHO-UMC

Naranjo Algorithm (Objective Scoring System)

- Evaluates 10 standard questions assigned numerical scores (e.g., +2, +1, 0, -1).
- Total scores categorize causality into:
    - **Definite:** >= 9
    - **Probable:** 5–8
    - **Possible:** 1–4
    - **Doubtful:** <= 0
- **Advantage:** High reproducibility and less subjective variation across reviewers.

WHO-UMC System (Expert Judgment / Broad Categories)

- Relies on qualitative clinical judgment using pre-defined criteria.
- Categories range from **Certain** down to **Probable**, **Possible**, **Unlikely**, **Conditional/Unclassified**, or **Unassessable/Unclassifiable** (when data is insufficient).
- **Advantage:** Highly versatile and universally used in global safety surveillance databases (like VigiBase).

3. Special Considerations

Adverse Events Following Immunization (AEFI)

- Uses dedicated algorithms (e.g., WHO AEFI Causality User Guide).

- Accounts for factors unique to biologicals, such as vaccine quality/cold-chain integrity, immunization process/program errors, timing of the event, cluster cases, and population-level background rates.

AI & Automation

- Natural Language Processing (NLP) and machine learning models are increasingly used to assist medical reviewers in triaging high-volume Individual Case Safety Reports (ICSRs), pre-populating causality scores, and identifying potential safety signals based on vast datasets of previous cases.